Assignment 10

CS 202 Lab Assignment

Total Marks =20 (Time Duration : 2 hours)

Q1. Develop a Java program for an online ticket booking system in which multiple users attempt to book tickets for a limited-capacity event. The system must use multiple threads to process booking requests and exception handling to manage invalid requests and ticket unavailability.

Requirements:

1. Create a class *TicketBookingSystem* that maintains the total number of available tickets.
2. Implement a synchronized method *bookTicket(String userName, int numberOfTickets)* that:
    a. Checks whether sufficient tickets are available. If not, throw a custom checked exception named *TicketsUnavailableException*.
    b. If tickets are available, reduce the available ticket count and display the booking details.
3. Create a class *BookingThread* that extends Thread. Each thread should represent a user attempting to book tickets.
4. Override the *run()* method to invoke *bookTicket()* and handle the exceptions appropriately.
5. In the main() method
    a. Initialize the system with 5 available tickets.
    b. Create at least five booking threads with different usernames and ticket requests.
    c. After all threads complete, display the total number of tickets remaining.